Farzaneh Derakhshan

RESEARCH INTERESTS

Formal Methods, Security and Privacy, Programming Languages, Verification, Type Theory.

PROFESSIONAL APPOINTMENTS

Northwestern University, **Research Assistant Professor**, Evanston, IL — October 2026 - present
Computer Science Department.

Illinois Institute of Technology, **Assistant Professor**, Chicago, IL — August 2023 - August 2026
Computer Science Department.

Carnegie Mellon University, **Postdoctoral Fellow**, Pittsburgh, PA — June 2021 - July 2023
Computer Science Department & CyLab Security and Privacy Institute.
Advisors: *Limin Jia, Stephanie Balzer.*

EDUCATION

Carnegie Mellon University, **Ph.D. in Pure and Applied Logic**, Pittsburgh, PA. Fall 2016 - Spring 2021
Advisor: *Frank Pfenning* Thesis: *Session-Typed Recursive Processes and Circular Proofs.*

Carnegie Mellon University, **M.Sc. in Computer Science–Research**, Pittsburgh, PA. Fall 2016 - Spring 2019
Advisor: *Frank Pfenning* Thesis: *Circular proofs as Session-Typed Processes: A Local Validity Algorithm.*

University of Tehran, **M.Sc. in Computer Science**, Tehran, Iran. Fall 2013 - Spring 2016
Thesis: *Uniform Interpolation in Substructural Logics.*

University of Tehran, **B.Sc. in Computer Science**, Tehran, Iran. Spring 2009 - Fall 2013
Dual-Degree Honors program. Thesis: *Martin-Lof's Intuitionistic Type Theory.*

Tehran University of Medical Sciences, **M.D., Medicine**, Tehran, Iran. Fall 2007 - Fall 2015
M.D. Thesis: *Design and implementation of a web-based CNS tumor registry in main hospitals of Tehran University of Medical Sciences.*

PUBLICATIONS

P. Azevedo de Amorim, **F. Derakhshan** *Quantitative Security for Session Types, Denotationally*, **ACM SIGPLAN Symposium on Principles of Programming Languages (POPL)** 2027. **Conditionally accepted.**

S. Balzer, **F. Derakhshan**, R. Harper, Y. Yao *Recursive Logical Relations for Intuitionistic Linear Logic Session Types*, **European Symposium on Programming (ESOP)** 2026.
Distinguished Paper Award and Best Theoretical Paper at ETAPS Award.

L. Liu, **F. Derakhshan**, L. Jia, G. Moreno, M. Klein, *Quantified Underapproximation via Labeled Bunches*, **Proceedings of the ACM on Programming Languages OOPSLA** 2025.

G. Moreno, M. Klein, S. Saha, **F. Derakhshan**, L. Jia,*Towards Compositional Assurance of Large Cyber-Physical Systems*, **Workshop on Formal Arguments for Cps CerTification (FACCT)** 2025.

M. Dotzel, **F. Derakhshan**, M. Surbatovich, L. Jia *Modal Crash Types for WAR-Aware Intermittent Computing*, **ACM Transactions on Programming Languages and Systems (TOPLAS)** 2025.

F. Derakhshan, S. Balzer, Y. Yao *Regrading Policies for Flexible Information Flow Control in Session-Typed Concurrency*, **European Conference on Object-Oriented Programming (ECOOP)** 2024.

B. van den Heuvel, **F. Derakhshan**, S. Balzer *Information Flow Control in Cyclic Process Networks*, **European Conference on Object-Oriented Programming (ECOOP)** 2024.

F. Derakhshan, M. Dotzel, M. Surbatovich, and L. Jia. *Modal crash types for intermittent computing.* 32nd **European Symposium on Programming (ESOP)** 2023.

F. Derakhshan, Z. Zhang, A. Vasudevan, and L. Jia. *Towards end-to-end verified TEEs via verified interface conformance and certified compilers.* **IEEE Computer Security Foundations Symposium (CSF)** 2023.

F. Derakhshan, F. Pfenning, *Circular proofs as Session-Typed Processes: A Local Validity Condition.* **Logical Methods in Computer Science (LMCS)** 18, 2022.

F. Derakhshan, S. Balzer, L. Jia, *Session Logical Relations for Noninterference.* 36th Annual ACM/IEEE Symposium on **Logic in Computer Science (LICS)**. Pages 1-14. IEEE 2021.

W. Sieg, **F. Derakhshan**, *Human-centered automated proof search.* **Journal of Automated Reasoning (JAR)** 65.8, 1153-1190, 2021.

M. Alizadeh, **F. Derakhshan**, H. Ono, *Uniform Interpolation in Substructural Logics.* **The Review of Symbolic Logic (RSL)**, Vol. 7, Issue 3, pp. 455 - 483, 2014.

PAPERS UNDER SUBMISSION

L. Liu, Z. Zhang, **F. Derakhshan**, L. Jia, W. Mansky, A. Vasudevan, *End-to-End Guarantees for Compartmentalized Safety Critical Software.* **CSF** 2027. **Major revision**

G. Bhogadi, **F. Derakhshan**, S. Muller, *A Relational Hoare Logic for GPU Kernel Side Channels.* 2027.

A. Madhu, L. Liu, S. Nawaz, **F. Derakhshan**, M. Zhou, *PIMSA: A Static Analysis Framework for Processing-in-Memory Programs.* **ASPLOS** 2027.

RESEARCH FUNDING & GRANTS

NSF SaTC CORE (Collaborative), Award #2350217 2024 - 2027
Mixed Assurance Reasoning via Modal Logic. Role: PI. Amount: $300,000.

SELECTED AWARDS

ETAPS Distinguished Paper Award 2026
For the paper *Recursive Logical Relations for Intuitionistic Linear Logic Session Types* (ESOP 2026).

EATCS Best Theoretical Paper at ETAPS 2026
For the same paper, selected as the best theoretical paper across the four ETAPS conferences (ESOP, FASE, FoSSaCS, TACAS) by an EATCS-appointed panel.

TEACHING

CS595. Language-based Security, Illinois Tech. Fall 2025

CS534. Types and Programming Languages, Illinois Tech. Spring 2025

CS440. Programming Languages and Translators, Illinois Tech. Fall 2024

CS536. Science of Programming, Illinois Tech. Fall 2023

80-210. Logic and Proofs, Carnegie Mellon University. Spring 2019, Summers 2017-2020

80-211. Logic and Mathematical Inquiry, Carnegie Mellon University. Fall 2018

ADVISING

Ph.D. Students

Lang Liu, Illinois Tech. Spring 2024 - present

Godha Pallavi Bhogadi, Illinois Tech (co-advised with Stefan Muller). Fall 2025 - present

Myra Dotzel, Carnegie Mellon University (co-advised with Limin Jia). Fall 2021 - Summer 2026
First position: Postdoctoral Research Fellow, University of Michigan.

M.Sc. Students

Akash Madhu, Illinois Tech. Spring 2025 - Spring 2026

Narasimha Karthik G, Illinois Tech (co-advised with Minxuan Zhou). Spring 2024 - Spring 2025
First position: Software Engineer, Amazon.

Undergraduate Researchers

Sidrah Nawaz, Illinois Tech. Summer 2025 - Spring 2026

Griffin Colomer, Illinois Tech. Spring 2025

PROFESSIONAL SERVICE

Organizer & Committee Chair

- **Organizing committee member** of Dagstuhl Seminar 26071 on Behavioural Types for Resilience (Feb. 8 - Feb. 13, 2026).
- **Program committee (co-)chair** of 16th Workshop on Programming Language Approaches to Concurrency & Communication cEntric Software (PLACES) 2025.

Program Committee Member & Reviewer

- *Conferences:* OOPSLA 2027, POPL 2027, LICS 2026, ESOP 2026, COORDINATION 2025, ICFP 2025, ECOOP 2025.
- Workshops: ICE 2025, PriSC 2025, TLLA 2024, PriSC 2024, APLAS 2022, ECOOP Doctoral Symposium 2022, PLACES 2022.
- Journals: ACM Transactions on Privacy and Security (TOPS) 2025.

Panels

- Panelist in PLMW 2021, Panel: Navigating PhD Studies.

Proposal/Thesis Committee Member

- Zichao Zhang, Carnegie Mellon University, 2025.
- Sajad Meisami, Illinois Tech, 2024. *Understanding Decentralized Applications From a Security Perspective.*

Departmental Service

- Co-Chair, CS Seminar Committee, Illinois Institute of Technology.
- Member, Graduate Studies Committee, Illinois Institute of Technology.
- Member, Faculty Hiring Committee, Illinois Institute of Technology.
- Member, PhD Experience & Recruiting Committee, Illinois Institute of Technology.

SELECTED TALKS

Mixed Assurance Reasoning via Labeled Bunches.. **F. Derakhshan**. **Association for Symbolic Logic North American Annual Meeting**. Special Session on Logic in Computer Science. Jul. 2026

Quantified Underapproximation via Labeled Bunches. **F. Derakhshan**. **(IFIP Working Group 2.1 on Algorithmic Languages and Calculi, Meeting #84).** Apr 2026

Modal Crash Types for Intermittent Computing. **F. Derakhshan**. **(IFIP Working Group 2.1 on Algorithmic Languages and Calculi, Meeting #82).** Nov. 2024

Modal Crash Types for Intermittent Computing. **F. Derakhshan**. **Workshop on Principles, Theory, and Practice for Decentralized Applications (PLAID 2024).** Sep. 2024

Session-Typed Recursive Processes and Infinitary Proofs. **F. Derakhshan**. **Special session on the semantics of non-wellfounded and circular proofs (MFPS 2024).** Jun. 2024

Towards End-to-End Verified TEEs via Verified Interface Conformance and Interface-Preserving Compilers. **F. Derakhshan**, Z. Zhang, A. Vasudevan, L. Jia. **7th Workshop on Principles of Secure Compilation (PriSC 2023).** Jan. 2023

Noninterference for session-typed processes. **F. Derakhshan**, S. Balzer, L. Jia. **16th Workshop on Programming Languages and Analysis for Security (PLAS 2021).** Dec. 2021

Infinitary proof theory of first order linear logic with fixed points. **F. Derakhshan**. **Association for Symbolic Logic North American Annual Meeting (online)**. Special Session on Proof Theory. Mar. 2020

Computational interpretation of substructural proofs. **F. Derakhshan**. **Association for Symbolic Logic North American Annual Meeting.** Logic and Philosophy Special Session. May 2018

SOFTWARE & ARTIFACTS

LabelBI proof-search tool. Interactive proof search for composing formally verified and tested component analyses; evaluated on a model of the Ariane-5 flight failure.
Artifact: L. Liu, **F. Derakhshan**, L. Jia, G. Moreno, M. Klein, *Quantified Underapproximation via Labeled Bunches* (OOPSLA 2025). doi:10.5281/zenodo.16929783

Information-flow type checker for session-typed concurrency. Type checker for an IFC language with regrading policies, with an interactive demonstration.
F. Derakhshan, S. Balzer, Y. Yao, *Regrading Policies for Flexible Information Flow Control in Session-Typed Concurrency* (Artifact). **Dagstuhl Artifacts Series (DARTS)** 10(2), 4:1–4:3, 2024. Artifact evaluated (ECOOP 2024). doi:10.4230/DARTS.10.2.4, zenodo.org/records/12735388

Circular proofs as session-typed processes. Implementation of the local validity condition for session-typed recursive processes.
A. Das, **F. Derakhshan**, F. Pfenning, v1.0, 2022. doi:10.5281/zenodo.6345260